PRE-ALGEBRA (7TH) · UNIT 4: FACTORS, FRACTIONS, AND EXPONENTS

Space Mission Packer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students pack a Mars mission. Divisibility, prime factorization, and the GCF create identical supply kits; fractions describe cargo space; exponents and scientific notation describe distances and signal times.

Every number skill in the unit has a job on the spaceship.

Standards Addressed

- 6.NS.B.4 GCF and LCM
- 8.EE.A.1 Properties of integer exponents
- 8.EE.A.3–4 Scientific notation
- 8.NS.A.1–2 Rational and irrational numbers

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a photo of a space station storage bay. Part 1 GCF and kits.
Day 2	Part 2 cargo fractions and real numbers.
Day 3	Part 3 exponents and scientific notation; build the manifest.

Materials

- Student handout and Supply Cards
- Calculators
- Optional: NASA Mars fact sheet

How the Three Versions Differ

Level	What changes
CP2	Divisibility rules, GCF of two numbers, simplifying fractions, and powers of 10.
CP1	Prime factorization with exponents, GCF of six numbers, fraction sums, rational vs. irrational, and exponent rules.
Honors	LCM schedules, fitting cargo with fractions, terminating vs. repeating decimals, negative and zero exponents.

Watch For (Common Misconceptions)

- Confusing GCF with LCM.
- Adding exponents when dividing powers.
- Writing 22.5×10^7 instead of 2.25×10^8.

Extension Ideas

- Research the real cargo of the International Space Station.
- Compute signal times to other planets.
- Design a kit for a moon base.

Answer Key – CP2

Part 1: Equal Kits

#	Task	Answer
1	Is 432 divisible by 2, 3, 6, and 9? Use divisibility rules.	Yes to all: even, digit sum 9 is divisible by 3 and 9, so also by 6.
2	Write the prime factorization of 96 using exponents.	$96 = 2^5 \times 3$
3	Find the GCF of 96 and 48. What does it tell you about packing kits?	GCF 48; you can make 48 identical kits, each with 2 oxygen canisters and 1 medical kit.
4	Find the GCF of 360 and 240. How many identical kits of meals and batteries can be made? What is in each?	GCF 120; 120 kits with 3 meal packs and 2 batteries each.

Part 2: Cargo Fractions

#	Task	Answer
5	Water fills 18 of the 24 cargo sections. Write this as a fraction in simplest form.	$\frac{18}{24} = \frac{3}{4}$
6	Food fills $\frac{12}{32}$ of the freezer. Simplify.	$\frac{3}{8}$
7	Which is larger, $\frac{3}{8}$ or $\frac{2}{5}$? Explain.	$\frac{3}{8} = 0.375$ and $\frac{2}{5} = 0.4$, so $\frac{2}{5}$ is larger.
8	Write $\frac{3}{4}$ as a decimal and a percent.	0.75 and 75%

Part 3: Space Numbers

#	Task	Answer
9	Write 10^8 in standard form. How many zeros does it have?	100,000,000; 8 zeros
10	Write 2.25×10^8 km in standard form.	225,000,000 km
11	The Moon is about 384,000 km away. Write this in scientific notation.	3.84×10^5 km

Reflection

#	Prompt	Look for
R1	Why does the GCF give the greatest number of identical kits?	It is the largest number that divides every amount evenly, so nothing is left over.

#	Prompt	Look for
R2	Why do scientists use scientific notation?	It makes very large or small numbers shorter and easier to compare.

Answer Key – CP1

Part 1: Supply Kits

#	Task	Answer
1	Write the prime factorization of 432, 360, and 144 using exponents.	$432 = 2^4 \times 3^3$; $360 = 2^3 \times 3^2 \times 5$; $144 = 2^4 \times 3^2$
2	Find the GCF of 432, 360, and 144. How many identical kits can be made with water, meals, and seeds? What is in each kit?	GCF = $2^3 \times 3^2 = 72$; 72 kits, each with 6 water, 5 meals, 2 seeds
3	Can all six supplies go into the same identical kits? Find the GCF of all six amounts.	GCF(432, 360, 96, 48, 144, 240) = 24 → 24 kits
4	List what goes in each of the 24 kits.	18 water, 15 meals, 4 oxygen, 2 medical, 6 seeds, 10 batteries

Part 2: Cargo Fractions

#	Task	Answer
5	Each kit fills $\frac{1}{3}$ of a section. How many sections do 24 kits fill? What fraction of the bay is that?	8 sections; $\frac{8}{24} = \frac{1}{3}$ of the bay
6	Science equipment fills $\frac{5}{12}$ of the bay and spare parts fill $\frac{1}{6}$. Write the fraction of the bay that is still empty.	$1 - \frac{1}{3} - \frac{5}{12} - \frac{1}{6} = \frac{12}{12} - \frac{4}{12} - \frac{5}{12} - \frac{2}{12} = \frac{1}{12}$
7	Classify each as rational or irrational: $\frac{5}{12}$, 0.333..., √2, π, √16.	Rational: $\frac{5}{12}$, 0.333..., √16 = 4. Irrational: √2, π.
8	Order $\frac{5}{12}$, 0.4, $\frac{3}{8}$, and $\frac{4}{9}$ from least to greatest.	$\frac{3}{8}$ (0.375), 0.4, $\frac{5}{12}$ (0.416...), $\frac{4}{9}$ (0.444...)

Part 3: Exponents in Space

#	Task	Answer
9	Simplify $10^8 \times 10^3$ and $\frac{10^8}{10^5}$. Explain the rules.	10^{11} (add exponents); 10^3 (subtract exponents)
10	Light travels about 3×10^5 km per second. About how many seconds does a radio message take to reach Mars at 2.25×10^8 km?	$\frac{2.25 \times 10^8}{3 \times 10^5} = 0.75 \times 10^3 = 750$ seconds = 12.5 minutes
11	Mars has a diameter of 6.8×10^3 km and Earth 1.27×10^4 km. About how many times wider is Earth?	$\frac{12{,}700}{6{,}800} \approx 1.9$ times

Reflection

#	Prompt	Look for
R1	How does prime factorization make finding the GCF faster?	You can take the shared primes with the smaller exponents instead of listing all factors.
R2	Explain why $\frac{10^8}{10^5} = 10^3$ using repeated multiplication.	Five of the eight tens cancel, leaving three tens multiplied together.

Answer Key – Honors

Part 1: Kits & Schedules

#	Task	Answer
1	Find the GCF of all six supply amounts using prime factorization and list one kit.	432 = $2^4 3^3$, 360 = $2^3 3^2 5$, 96 = $2^5 3$, 48 = $2^4 3$, 144 = $2^4 3^2$, 240 = $2^4 3 \cdot 5$ → GCF $2^3 \times 3 = 24$
2	Oxygen is checked every 6 days, batteries every 8 days, and seeds every 15 days. All three are checked on day 0. When are they next checked on the same day?	LCM(6, 8, 15) = 120 → day 120
3	If the mission lasts 540 days, how many times are all three checked on the same day (including day 0)?	0, 120, 240, 360, 480 → 5 times
4	A second ship carries 504 water pouches. Can it use the same kits (with 18 water each)? Explain with factors.	504 ÷ 18 = 28 kits; yes, 18 is a factor of 504 (504 = $2^3 \times 3^2 \times 7$).

Part 2: Cargo & Real Numbers

#	Task	Answer
5	Kits fill $\frac{1}{3}$, science $\frac{5}{12}$, and spare parts $\frac{1}{6}$ of the bay. The crew wants to add a rover that fills $\frac{3}{16}$ of the bay. Will it fit? Show your work.	Empty = $\frac{1}{12} \approx 0.083 < \frac{3}{16} = 0.1875$, so it will not fit.
6	How much would the science equipment need to shrink (as a fraction of the bay) so the rover fits?	$\frac{3}{16} - \frac{1}{12} = \frac{9}{48} - \frac{4}{48} = \frac{5}{48}$ of the bay
7	Between which two whole numbers is √50? Is it rational? Use this to estimate the side of a square panel with area 50 m².	7 and 8 (√49, √64); irrational; about 7.07 m
8	The radiation shield is $\frac{4}{9}$ lead. Is the decimal terminating or repeating? Explain using the prime factors of the denominator.	Repeating (0.444…): the denominator 9 = 3^2 has a prime factor other than 2 or 5.

Part 3: Powers of Space

#	Task	Answer
9	Simplify and evaluate: $\frac{(2^3)^2 \times 2^{-4}}{2^0}$.	$2^6 \times 2^{-4} \div 1 = 2^2 = 4$

#	Task	Answer
10	At the closest approach Mars is 5.46×10^7 km away; at the farthest it is 4.01×10^8 km. How many times farther is the farthest distance? How long does a radio signal take for each (light travels 3×10^5 km/s)?	$\frac{4.01 \times 10^8}{5.46 \times 10^7}$ ≈ 7.3 times. Signals: 182 s (about 3 min) and about 1,337 s (about 22 min).
11	A dust particle has a mass of 5×10^{-7} kg. How many particles make 1 kg? Write your answer in scientific notation.	$\frac{1}{5 \times 10^{-7}}$ = 2×10^6 particles

Reflection

#	Prompt	Look for
R1	When is the GCF the right tool and when is the LCM? Give mission examples.	GCF for splitting supplies into equal kits; LCM for finding when repeating schedules line up.
R2	Why is $a^0 = 1$ for any nonzero a? Use the quotient rule.	$\frac{a^n}{a^n}$ = 1 and also = $a^{n-n} = a^0$, so $a^0 = 1$.